

CARBON FOOTPRINT REPORT

GIGS Manufacturing Pvt. Ltd.

Reporting Period: FY 2025–26 (April 2025 – March 2026)

Total GHG Emissions

341.44 t CO₂e

Scope 1: 26.6% | Scope 2: 50.3% | Scope 3: 23.1%

Prepared in line with the GHG Protocol Corporate Standard

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Prepared as part of the MAESA Foundation carbon footprint training exercise

1. Executive Summary

GIGS Manufacturing Pvt. Ltd. emitted 341.44 t CO₂e in FY 2025–26. Purchased electricity (Scope 2) is the largest contributor at 50.3% of the total, followed by direct emissions (Scope 1) at 26.6% and other indirect emissions (Scope 3) at 23.1%.

Four sources (grid electricity, purchased raw materials, diesel and refrigerant leakage) together make up about 90% of the footprint. Reduction efforts should therefore focus on renewable electricity and efficiency, diesel displacement, refrigerant leak control and supplier engagement.

Scope	Emissions (t CO ₂ e)	% of total
Scope 1: Direct	90.70	26.56%
Scope 2: Purchased energy	171.84	50.33%
Scope 3: Other indirect	78.90	23.11%
Total	341.44	100.00%

GIGS Manufacturing Pvt. Ltd. — GHG Emissions by Scope
FY 2025–26

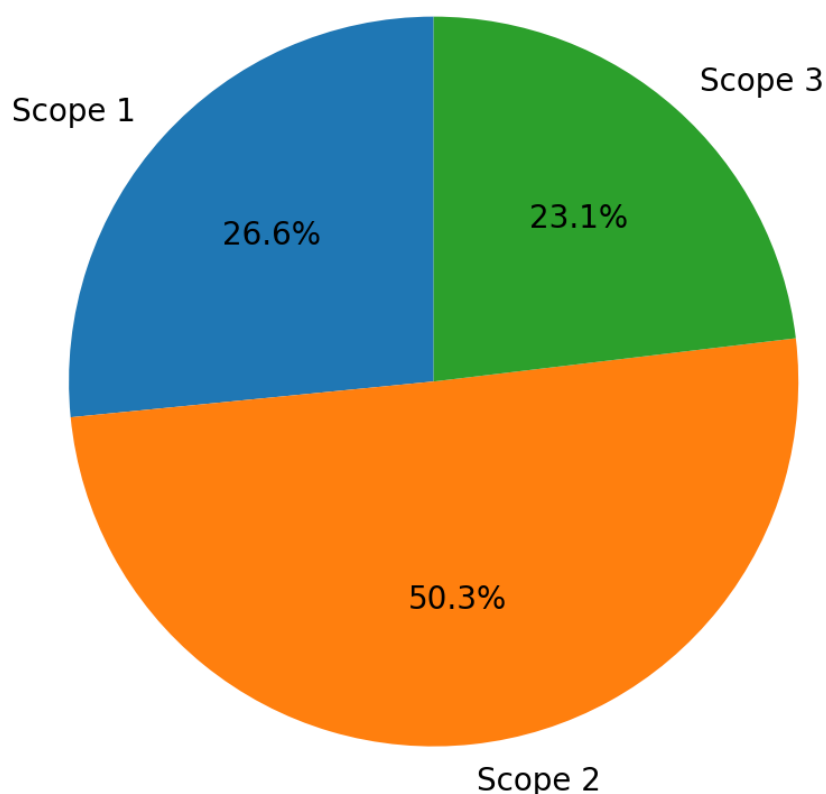


Figure 1: GIGS Manufacturing Pvt. Ltd. — Share of total emissions by scope

2. Organizational and Reporting Boundary

- **Reporting entity:** GIGS Manufacturing Pvt. Ltd., a small manufacturing company.

- **Operational boundary:** one manufacturing facility and one administrative office.
- **Consolidation approach:** operational control.
- **Reporting period:** FY 2025–26 (April 2025 to March 2026).
- **Standard and gases:** GHG Protocol Corporate Standard; emissions reported in CO₂e.
- **Scope 3 coverage:** purchased goods (raw materials), employee commuting and waste generated. Other categories, such as upstream transportation, business travel and use of sold products, are excluded due to lack of data.

3. Activity Data

Scope	Activity	Quantity
1	Diesel consumed in generators and company-owned vehicles	18,000 litres/year
1	Petrol consumed in company-owned vehicles	6,000 litres/year
1	Refrigerant leakage from air-conditioning systems	20 kg/year
2	Electricity purchased from grid	240,000 kWh/year
3	Employee commuting using personal vehicles	80,000 km/year
3	Raw materials purchased from suppliers	120 tonnes/year
3	Non-hazardous waste sent for external treatment	15 tonnes/year

4. Emission Factors and Methodology

Emissions were calculated using the activity-data × emission-factor approach:

$$\text{Emissions} = \text{Activity Data} \times \text{Emission Factor}$$

Results in kilograms were divided by 1,000 to give tonnes CO₂e and rounded to two decimals. Scope 2 follows the location-based method using the grid factor supplied for this exercise.

Activity	Emission factor	Scope
Diesel	2.68 kg CO ₂ /litre	1
Petrol	2.31 kg CO ₂ /litre	1
Refrigerant (leakage)	1,430 kg CO ₂ e/kg	1
Electricity (grid)	0.716 kg CO ₂ /kWh	2
Employee commuting	0.18 kg CO ₂ e/km	3
Purchased raw materials	0.50 kg CO ₂ e/kg	3
Waste treatment	0.30 kg CO ₂ e/kg	3

5. Emission Inventory (Scope 1, 2 and 3)

Activity classification and calculations

#	Activity	Scope	Calculation	t CO ₂ e
1	Diesel	1	$18,000 \times 2.68$	48.24
2	Petrol	1	$6,000 \times 2.31$	13.86
3	Refrigerant leakage	1	$20 \times 1,430$	28.60
4	Grid electricity	2	$240,000 \times 0.716$	171.84
5	Employee commuting	3	$80,000 \times 0.18$	14.40
6	Purchased raw materials	3	$120,000 \text{ kg} \times 0.50$	60.00
7	Waste treatment	3	$15,000 \text{ kg} \times 0.30$	4.50
	Total			341.44

Scope-wise inventory

Scope	Source	t CO ₂ e	% of total
1	Diesel	48.24	14.13%
1	Petrol	13.86	4.06%
1	Refrigerant	28.60	8.38%
	Scope 1 total	90.70	26.56%
2	Electricity	171.84	50.33%
	Scope 2 total	171.84	50.33%
3	Commuting	14.40	4.22%
3	Raw materials	60.00	17.57%
3	Waste	4.50	1.32%
	Scope 3 total	78.90	23.11%
	Grand total	341.44	100.00%

6. Results and Key Findings

Major sources of emissions (ranked)

Rank	Source	Scope	t CO ₂ e	% of total
1	Grid electricity	2	171.84	50.3%
2	Purchased raw materials	3	60.00	17.6%
3	Diesel	1	48.24	14.1%
4	Refrigerant leakage	1	28.60	8.4%
5	Employee commuting	3	14.40	4.2%
6	Petrol	1	13.86	4.1%
7	Waste treatment	3	4.50	1.3%

Key findings

1. The total footprint is 341.44 t CO₂e, split 26.6% Scope 1, 50.3% Scope 2 and 23.1% Scope 3.
2. Grid electricity is the single largest hotspot at 171.84 t (50.3%).
3. The top four sources account for about 90% of total emissions.
4. Diesel is the largest Scope 1 source at 48.24 t (53% of Scope 1), so on-site diesel generation is a clear opportunity.
5. Refrigerant leakage is only 20 kg but contributes 28.6 t because of its high global warming potential.
6. Raw materials are the largest Scope 3 source at 60 t (76% of Scope 3).
7. Waste treatment is minor at 1.3% of the total.

7. Recommendations

1. **Reduce electricity emissions:** install rooftop solar or buy renewable power (green open access or a PPA); every 10% of grid electricity replaced saves about 17 t CO₂e. Improve efficiency with LED lighting, VFDs, efficient motors and an energy audit.
2. **Reduce diesel use:** rely more on grid or solar power, maintain and right-size generators, and shift company vehicles to electric or CNG.
3. **Control refrigerant leakage:** run leak-detection and preventive maintenance programmes, and switch to lower-GWP refrigerants at replacement.
4. **Address raw materials:** engage suppliers, source recycled or lower-carbon materials, cut material waste and collect supplier-specific emission data.
5. **Employee commuting:** offer shuttle buses, carpooling and public transit incentives, and hybrid or remote work where feasible.
6. **Waste:** segregate waste, recycle and reuse, and divert waste from disposal.
7. **Improve the inventory:** track data monthly, expand Scope 3 coverage, set FY 2025–26 as the baseline year, adopt a reduction target (for example 20–30% by FY 2030), and report intensity metrics such as t CO₂e per tonne of product.